

PKTQ: Per-Kernel Thresholded Quantization With Size-Constrained Two-Stage Optimization for CNNs

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ABSTRACT

Convolutional neural networks are challenging to deploy on resource-constrained computing platforms due to their high storage and computational demands. Mixed-precision quantization can mitigate these issues by assigning different bit-widths per layer. However, finding an optimal quantization policy is challenging due to the exponential size of the policy search space and the need for extensive policy evaluations. Additionally, existing works often overlook strict model size constraints and apply per-layer quantization, which ignores intra-kernel variances. We propose a two-stage deep deterministic policy gradient (DDPG) framework. In the first stage, it optimizes accuracy; in the second, it jointly optimizes accuracy and model size with a constraint-aware reward that enforces storage limits during training. On this basis, a per-kernel clipping threshold is introduced to reduce quantization error by adapting the clipping threshold to individual kernel distributions. We perform extensive simulations based on ImageNet with MobileNet-V1/V2 and ResNet-50. The results show that our method outperforms the state-of-the-art baselines under the same compression ratios, achieving minimal accuracy drops with aggressive compression and near-lossless performance at moderate levels.

1 | Introduction

Convolutional neural networks (CNNs), are widely used in multiple scenarios such as computer vision tasks and recommendation systems. However, their deployment in resource-constrained environments remains challenging due to high storage requirements and computational demands. For example, deploying a full-precision model like VGG16 (528 MB) [1] on a smartphone SoC is challenging because limited on-chip memory cannot support its size, which leads to frequent DRAM accesses that increase latency and power consumption.

Quantization technology can mitigate storage and computation bottlenecks by reducing both the model size and the required memory bandwidth, where the full-precision floating-point weights can be represented by reduced-precision integers [2].

To perform CNN quantization, some methods use the same bit-width for all layers, such as [3, 4]. However, a uniform bit-width allocation across all layers ignores the differences among layers, for example, the layer structure. Therefore, uniform bit-width will cause more information loss in the critical layers, leading to significant degradation in model accuracy.

Mixed-precision quantization technology selects an appropriate bit-width for each layer of a CNN, providing system flexibility and adaptivity, compared with fixed-precision quantization. Therefore, mixed-precision quantization can take the characteristics and behaviour of different layers into account [5], when a quantized CNN is applied to a hardware platform. On the other hand, with the recent advances in embedded AI hardware, many platforms support mixed-precision operations, such as the NVIDIA Jetson edge computing platform, which provides

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